

DOKULA MOHIT KUMAR

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Summary

Electronics and Communication Engineering graduate with a strong foundation in embedded systems, embedded C, and electronics. Hands-on experience in sensor interfacing, monitoring, and automation projects using microcontrollers. Passionate about embedded software development and eager to apply technical skills while continuously learning new technologies in the embedded systems domain.

Education

Bachelor of Technology (B.Tech) – (ECE) <i>Dr. Lankapalli Bullaya College of Engineering</i> CGPA : 8.06	2022 – 2026 Visakhapatnam
Intermediate(MPC) <i>Sri Chaitanya Junior College</i> Percentage: 94%	2020 – 2022 Visakhapatnam
Secondary School Certificate (SSC) <i>Marvel School</i> Percentage: 98%	2020 Visakhapatnam

Skills

Programming Languages:

C, Embedded C

Microcontrollers:

STM32 (ARM Cortex-M), ESP32, Arduino

Embedded Interfaces & Peripherals:

GPIO,UART, SPI, I2C,ADC/DAC, Timers/PWM, Sensor & Actuator Interfacing

Cloud Platforms:

Rightech IoT Cloud

Development Tools:

STM32CubeIDE, Arduino IDE, KiCad (PCB Design),Zephyr SDK

Operating Systems:

Linux (Ubuntu)

Embedded Operating Systems

Zephyr RTOS

Certificates

NPTEL Certificates

- Design Design With Verilog
- Microsensors and Nanosensors

BSNL

- Completed Telecommunications

BSNL Internship

- Successfully Completed Internship for Two Weeks in **TELECOM**

Projects

Industry Parameters Monitoring 4.0

Technologies: STM32CubeIDE, Embedded C, UART, I2C, ADC, MQTT, IoT Cloud

- Developed an IoT-based industrial monitoring system using **STM32F446RE** to track temperature, gas, vibration, pressure, and flame parameters in real time.
- Integrated multiple sensors using **ADC, GPIO, and I2C** communication protocols.
- Configured and interfaced **WE10 Wi-Fi module** using **UART** for cloud connectivity.
- Implemented real-time data transmission to **Right Tech Cloud** using MQTT/dashboard monitoring.
- Enabled local LCD display and remote cloud-based visualization for industrial monitoring.

IoT-Based Smart Parking System

Technologies: STM32CubeIDE, Embedded C, UART, SPI, I2C, PWM, HTTPS/REST, IoT, Cloud Database, Frontend

- Developed an IoT-based Project using **STM32F446RE** with RFID-based access and automated gate control.
- Interfaced peripherals using **SPI, I2C, GPIO, and PWM**, and integrated a **Wi-Fi module over UART** for secure **HTTPS/REST** communication with a cloud database.
- Implemented RFID authentication, parking fee calculation, real-time parking status updates, and reliable synchronization after system resets.
- Designed a **web-based admin dashboard** to monitor parking status, slot availability, registered users, and parking earnings in real time.

Distance Monitoring System

Technologies: STM32CubeIDE, Embedded C, UART, Timer Input Capture, MQTT, Right Tech IoT Cloud

- Developed an IoT-based distance monitoring system using **STM32F446RE** and the **HC-SR04 ultrasonic sensor** for real-time distance measurement.
- Implemented **Timer Input Capture** to accurately calculate distance and interfaced the **WE10 Wi-Fi module** via **UART** for wireless communication.
- Transmitted real-time distance data to **Right Tech IoT Cloud** using the **MQTT protocol** for remote monitoring and visualization.

Environmental Monitoring System using Zephyr RTOS

Technologies: Zephyr RTOS, Embedded C, STM32 Nucleo-F446RE, BME280, I2C, UART, WE-10, RightTech IoT Cloud

- Developed an environmental monitoring application using **Zephyr RTOS** on the **STM32 Nucleo-F446RE** board by interfacing the **BME280** sensor over **I2C**.
- Configured hardware using **DeviceTree overlays** and implemented sensor data acquisition using Zephyr device drivers.
- Transmitted real-time **temperature, humidity, and pressure** data to **RightTech IoT Cloud** through the **WE-10 Wi-Fi module** using **UART**.

Personal Profile

Father:

Dokula PoliNaidu

Mother:

Dokula Rupa Vathi

Date of Birth:

12-06-2005

Address:

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